Roll Number: _____

Thapar Institute of Engineering and Technology, Patiala

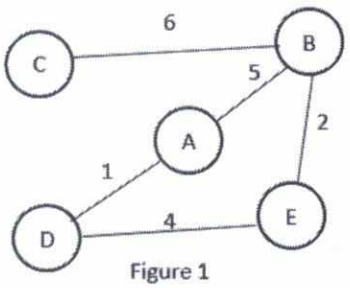
Department of Computer Science and Engineering

EST Summer Examination (Summer Semester-2025)

B.E.
24th July 2025
Duration: 3 Hours
Max Marks: 40

Course Code: UCS414/UCS520
Course Name: Computer Networks
Name of Faculty: Dr. Sumit Kumar

Note: Attempt any five questions. Assume missing data, if any, suitably.

Q1	(a) List the seven layers of the OSI reference model and explain the primary function of the network layer and presentation layer in the data communication process.	(4)
	(b) A synchronous time division multiplexer combines four 500-kbps channels and multiplexes 1 bit per channel. Draw a suitable diagram to show input data and output frames with arbitrary inputs. Compute the frame rate, frame duration, and bit duration after multiplexing.	(4)
Q2	(a) Assume that a computer network employs a Dynamic Host Configuration Protocol (DHCP) server for the dynamic assignment of IP addresses to devices connected to the network. Suppose a new device is added to this network and requires an IP address. Describe the sequence of operations performed by the DHCP protocol that enable the device to successfully obtain a valid IP address from the server and become a functioning part of the network communication.	(4)
	(b) Given an IPv4 address 192.168.10.50 with CIDR notation /26, calculate the corresponding network address and broadcast address.	(4)
Q3	(a) Consider a network with five nodes, A to E, as shown in Figure 1. The figure defines the cost to traverse from one node to another on the connections. The network uses the link-state routing algorithm. Prepare the initial link state packets created by each node in the network. Suppose node "B" received the initial link-state packet from every other node in the network, then prepare the link-state database on node "B".  Figure 1	(4)

	(b) Determine the Hamming code to be transmitted for a 4-bit data 1011 by identifying and placing the parity bits (assume even parity) at appropriate positions.	
Q4	(a) Compare Go-Back-N ARQ and Selective Repeat ARQ protocols. Highlight the differences in terms of efficiency, buffer utilization, and retransmission strategy. (b) Discuss the following network congestion control approaches with a suitable diagram: (i) Back pressure, and (ii) Choke packet.	(4) (4)
Q5	(a) Explain the key features of SCTP. Describe the role of the following fields in SCTP operations: Transmission Sequence Number (TSN), Stream Identifier (SI), and Stream Sequence Number (SSN). (b) Discuss the differences between the following access control mechanisms: Carrier Sense Multiple Access with Collision Detection (CSMA/CD) and Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA).	(4) (4)
Q6	(a) Frequency Division Multiple Access (FDMA) and Code Division Multiple Access (CDMA) are two important multiple access techniques used in communication systems. Compare and contrast FDMA and CDMA with a suitable example. (b) The IEEE 802.3 standard defines multiple physical layer Ethernet implementations, including 10Base2, 10Base5, and 10BaseT. Differentiate among these three Ethernet standards.	(4) (4)

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